



RB001 Week 5 — English Worksheet

Topic: Loops — Platforms that Vanish · **Course:** Obby Architect · **Time:** about 45 minutes

This week your scripts **repeat** using loops. A **for** loop repeats a set number of times. A **while true do** loop repeats forever. With loops, one platform can fade out, vanish, and come back — again and again.

1 · Predict 🧠

Read each script. Before you imagine running it in Studio, write what you think will happen.

```
local platform = script.Parent

for i = 1, 10 do
    platform.Transparency = i / 10
    task.wait(0.1)
end
```

The loop runs 10 times. What does the platform look like when the loop ends? About how many seconds does the fade take?


```
local platform = script.Parent

while true do
    task.wait(3)
    platform.Transparency = 1
    platform.CanCollide = false
    task.wait(2)
    platform.Transparency = 0
    platform.CanCollide = true
end
```

This loop never stops. What does the platform do, over and over? For how many seconds is it safe to stand on?


```
local platform = script.Parent

while true do
    for i = 1, 10 do
        platform.Transparency = i / 10
        task.wait(0.1)
    end
    platform.CanCollide = false
    task.wait(2)
    platform.Transparency = 0
    platform.CanCollide = true
    task.wait(3)
end
```

You are standing on this platform when the fade finishes. What happens to you? When is it safe to step on again?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the script is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The platform should blink: vanish, wait 2 seconds, come back, wait 2 seconds, forever. But when you press Play, Studio **freezes**.

```
local platform = script.Parent

while true do
    platform.Transparency = 1
    platform.CanCollide = false
    platform.Transparency = 0
    platform.CanCollide = true
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The platform is invisible. This loop should fade it **back in**, from invisible to solid. But nothing happens at all.

```
local platform = script.Parent

for i = 10, 1 do
    platform.Transparency = i / 10
    task.wait(0.1)
end
```

Hint: counting **down** needs one extra number in the **for** line.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The platform should fade out, vanish for 2 seconds, then come back good as new. But after the first cycle, players fall through a platform they **cannot even see** — and it never looks solid again.

```
local platform = script.Parent

while true do
    for i = 1, 10 do
        platform.Transparency = i / 10
        task.wait(0.1)
    end
    platform.CanCollide = false
    task.wait(2)
    platform.CanCollide = true
    task.wait(3)
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your obby in Roblox Studio. Build a **vanishing bridge**: at least 3 platforms that fade out and come back using a loop. Cross it yourself in Play mode. When you finish, come back here.

Send a photo or video of your bridge while it is fading, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I used a **loop** to ...

The platform fades because `Transparency = i / 10` means ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you cross your vanishing bridge in Play mode. Talk through it like you are teaching someone who has never seen it. Try to use these words: **loop, repeat, forever, fade, wait.**

1. Show your bridge, then press Play.
2. Read your `while true do` loop out loud and say what repeats.
3. Point at `task.wait` and say what would happen without it.
4. Cross the bridge and say when it is safe and when it is not.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.